

CHT Content Hub KOL + HCP Intel Migration Spec

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Backend + data migration from chm-mediahub
Frontend rebuild - API contracts only

Source: docs/kol-hcp-intel-migration.md

Target: zero KOL and HCP Intel ownership in MediaHub after cutover.

KOL + HCP Intel migration spec

Status: Approved target state - June 2026

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Scope: Everything KOL- and HCP-Intel-related moves from chm-mediahub into cht-content-hub at the backend + data layer. MediaHub retains zero ownership of KOL data, static assets, APIs, or HCP Intel after cutover. Frontend is rebuilt separately against Content Hub APIs.

Supersedes interim language in contenthub-admin-architecture.md decision #5 ("backed by MediaHub RDS until a future HCP intel migration"). HCP Intel migrates with KOL in the same cutover - not as a follow-on.

Related: contenthub-migration-plan.md · cht-public-api-contract.md · contenthub-admin-architecture.md

Frontend: Consumer and admin UIs are being redone from scratch. This migration delivers backend + data + API contracts only. No React/Next.js pages, components, or E2E specs are ported from MediaHub or cht-platform-tool.

1. Final-state principle

System	After cutover
cht-content-hub	Single system of record: KOL + HCP Intel tables, public producer API, admin studio API, static headshots (CDN), ingestion jobs
New frontend (consumer + admin)	Built against Content Hub APIs. Legacy UIs in MediaHub and cht-platform-tool are reference only - not ported.
chm-mediahub	Client analytics, webhooks, report generator (if retained). No KOL routes, tables, static files, or HCP Intel. May read KOL/HCP counts via Content Hub API for dashboards.

contenthub.communityhealth.media	
---- /kol-network/*	New consumer UI (rebuild)
---- /admin/studio/kols/*	New admin UI (rebuild)
---- /admin/studio/hcp-intel/*	New admin UI (rebuild)
---- /api/public/kols/*	Producer (X-API-Key) - server-to-server + optional BFF
---- /api/public/hcp/upsert	CHT registration sync (X-API-Key)
---- /api/admin/studio/kols/*	KOL CMS + group management (Cognito JWT)
---- /api/admin/studio/hcp-intel/*	HCP Intel admin API (Cognito JWT)
`---- CDN /assets/kols/*	Headshots + report photos
mediahub.communityhealth.media	(no KOL, no HCP Intel)
---- /dashboard/*	Analytics (optional read-only Content Hub calls)
---- /webhook/sync	Ingest pipeline
`---- /api/...	Reports, client metrics - zero KOL/HCP Intel routes

1.1 Frontend redo - what changes vs what stays

In scope (migrate)	Out of scope (do not port)
Public producer API + schemas	MediaHub /dashboard/hcps/* pages (12)
Admin studio API (~60 routes)	MediaHub KOL group dashboard pages
Database tables + data dump	KOLGroupCard, KOLChip, HCP Intel components
Lambda / ingestion jobs	cht-platform-tool DolNetwork, KolProfilePage, DolRegionDetail
Headshots -> S3/CDN	dol-network.ts static mock + useKolDirectory merge logic
API contract docs (handoff to frontend team)	E2E specs (hcp-intel/*.spec.ts)
CHT hcp/upsert server path (registration)	cht-platform-tool kol-network proxy module (optional - drop if new frontend calls Content Hub directly)

Handoff artifact for frontend: OpenAPI or typed schema exports from Content Hub plus the route map in Section 3 and enrichment field list in Section 6.1. Existing pages are behavioral reference (what screens exist, what data they show) - not copy sources.

2. Is HCP Intel required? (Recommendation: yes - migrate wholesale)

HCP Intel is not optional if KOL network is in scope. The two domains are tightly coupled:

Coupling	Why it blocks a KOL-only migration
Publications	GET /api/public/kols/{slug}/publications reads hcp_signals via kols.hcp_npi
KOL <-> HCP link	kol_hcp_matcher.py resolves kols.hcp_npi -> hcps.npi
Rankings	Global KOL rankings and CHM engagement tiers are computed entirely from HCP Intel tables (rankings.py, webinar attendance, Rx volumes, NCI designations)
CHT registration	POST /api/public/hcp/upsert fires on every CHT signup/profile update - populates hcps that KOL matcher and enrichment depend on
Profile enrichment fields	Fields previously mocked in legacy dol-network.ts (NPI, AI brief, Open Payments, social URLs) must be served by the API from HCP Intel tables (hcp_ai_briefs, open_payments_records, hcps.enrichment_data) before the new frontend ships
Admin workflows	Review queue, NPI disambiguation, OpenAlex backfill, and data-quality tooling are all HCP Intel - needed to keep KOL profiles accurate

Verdict: Migrate HCP Intel with KOL in one cutover. Splitting them leaves publications broken, rankings orphaned, and admin staff bouncing between two systems.

What HCP Intel provides (business value)

Capability	Primary consumers
HCP roster + NPI resolution	Admin directory, CHT registration sync
Signal timeline (pubs, trials, news, Rx shifts)	HCP profile pages, KOL profile enrichment

OpenAlex publications	Public /kol-network publications tab
Webinar attendance + CHM engagement	CHM KOL rankings, prescriber impact
CMS Part D / Rx volumes	Prescriber impact, global KOL rankings
Open Payments + NIH grants	HCP profile detail, AI briefs
Review queue + data quality	Ops team daily workflow
AI briefs (Claude Haiku)	Profile enrichment, future public KOL detail

3. Source inventory (chm-mediahub -> cht-content-hub)

3.1 Public producer API

Port from chm-mediahub/backend/src/:

Endpoint	Handler	Module
GET /api/public/kols	get_public_kols	public/router.py
GET /api/public/kols/{slug}	get_public_kol_detail	public/router.py
GET /api/public/kols/{slug}/publications	get_public_kol_publications	public/router.py
POST /api/public/hcp/upsert	hcp_upsert	public/router.py + services/hcp_upsert.py

Supporting modules:

File	Purpose
models/kol.py	KOL, KOLGroup, KOLGroupMember
schemas/public.py	PublicKOL*, PublicKOLPublication*, HCPUpsert*
utils/public.py	kol_slug, kol_to_public, build_kol_slug_map
services/kol_regions.py	Region taxonomy + institution inference
hcp_intel/models.py	All HCP Intel SQLAlchemy models
hcp_intel/kol_hcp_matcher.py	KOL -> NPI matching
hcp_intel/rankings.py	Global + CHM KOL rankings
hcp_intel/openalex_backfill.py	Publication ingestion
hcp_intel/orchestrator.py	Feed polling orchestration
hcp_intel/* (full package)	Ingestion, disambiguation, AI brief, webinar, Rx, etc.

Legacy shim to retire after cutover: legacy/routers/public_api.py.

3.2 Admin studio API

Port from chm-mediahub/backend/legacy/routers/hcp_intel.py (~60 routes) under /api/admin/studio/hcp-intel/*:

Route group	Examples
Roster	GET/POST /hcps, GET/PATCH /hcps/{npi}, promote, timeline
Discovery + review	/discovery, /review/queue, /review/{id}
Rankings	/rankings/global, /rankings/chm, /kols roster
Analytics	/overview, /geo/states, /prescriber-impact, /cohort-flow
Webinars	/webinars, attendance, drugs, JotForm webhook
Medications	/medications, /manufacturers, /drugs/search
Per-HCP detail	NIH grants, Open Payments, AI brief, Rx, engagement, news, shifts
Admin ops	poll, OpenAlex backfill, signal purge, data-quality, ensure-plumbing

Also port KOL group admin from legacy/routers/clients.py (KOL-specific routes only) under /api/admin/studio/kols/*:

Route	Purpose
GET /kols	Global KOL list
Project KOL group CRUD	Groups, members, stats (currently under /api/clients/...)

3.3 Admin UI - reference only (new frontend rebuild)

Do not port MediaHub or cht-platform-tool frontend code. Use legacy pages as a feature checklist for the new admin SPA. Each row maps a screen the new UI should cover to the studio API routes it consumes.

HCP Intel screens (legacy: chm-mediahub/frontend/src/app/dashboard/hcps/):

Screen (new route TBD)	Studio API consumed	Legacy reference
Overview	GET /overview, /geo/states	/dashboard/hcps
HCP directory	GET /hcps	/dashboard/hcps/directory
HCP profile	GET /hcps/{npi}, timeline, grants, payments, ai-brief, rx, engagement	/dashboard/hcps/[npi]
Global KOL rankings	GET /rankings/global	/dashboard/hcps/kol-rankings
CHM KOL rankings	GET /rankings/chm	/dashboard/hcps/chm-kol-rankings
Prescriber impact	GET /prescriber-impact	/dashboard/hcps/prescriber-impact
Medications	GET /medications, /manufacturers	/dashboard/hcps/medications
Webinars	GET/POST /webinars, drugs, forms	/dashboard/hcps/webinars
Surveys	GET /surveys	/dashboard/hcps/surveys
Review queue	GET /review/queue, POST /review/{id}	/dashboard/hcps/review
Data quality	GET /admin/data-quality/*	/dashboard/hcps/data-quality

KOL admin screens (legacy: client dashboard):

Screen (new route TBD)	Studio API consumed	Legacy reference
KOL roster	GET /kols	-
Project KOL groups	project + group list endpoints	/dashboard/clients/.../projects/...
KOL group detail	group detail + members + clips	/dashboard/clients/.../groups/[groupId]

3.4 Static assets

Asset	Source path	Count	Target
KOL headshots	backend/legacy/static/kol-headshots/	29 PNG	S3 + CloudFront; rewrite kols.photo_url
Report KOL photos	backend/legacy/report_automation/assets/kol_photos/	~15 JPG	Same bucket; report generator reads from Content Hub CDN

Retire MediaHub mount: legacy/main.py -> StaticFiles(directory=_static_dir).

3.5 Legacy frontend - retire, do not migrate

These files in cht-platform-tool and MediaHub are superseded by the frontend redo. No action during backend migration except ensuring API fields exist before the new UI launches.

Legacy file	Disposition
cht-platform-tool/.../dol-network.ts	Retire - enrichment must come from API
cht-platform-tool/.../useKolDirectory.ts	Retire
cht-platform-tool/.../DoINetwork.tsx, KolProfilePage.tsx, DoIRegionDetail.tsx	Retire
chm-mediahub/frontend/.../dashboard/hcps/*	Retire after new admin UI ships
chm-mediahub/frontend/.../KOLGroupCard.tsx, KOLChip.tsx	Retire

4. Database tables

KOL (required)

Table	Purpose
kols	KOL profiles + hcp_npi bridge
kol_groups	Shoot groupings
kol_group_members	KOL <-> group links

Also required for shoot stats on public KOL API: shoots (via kol_group_id).

HCP Intel (required - migrate with KOL)

Table	Purpose
hcps	NPI-keyed identity; CHT upsert target
feed_sources	Source registry (PubMed, OpenAlex, etc.)
feed_subscriptions	Per-HCP polling state
feed_items	Raw ingested items
hcp_signals	Derived facts (publications, trials, news, ...)
signal_drugs	Drug linkage on signals
webinar_events	CHM webinar catalog
webinar_attendance	CHM engagement (CHM rankings)
webinar_drugs	Drug tags on webinars
webinar_forms	JotForm bindings
unmatched_attendees	Attendee resolution queue
rx_volumes	CMS Part D prescriber data
rx_drug_aliases	Drug name normalization
drug_classes	Therapeutic class taxonomy
drug_to_class	Drug <-> class mapping
medications	Medication catalog
manufacturers	Manufacturer catalog
open_payments_records	Sunshine Act data
hcp_nih_grants	NIH grant data
hcp_ai_briefs	Claude-generated profile briefs
nci_designations	NCI center prestige (global rankings)
data_sync_state	Ingestion job health

Alembic migrations to port (from chm-mediahub)

KOL-specific:

- 7c596193eb7d_add_multi_tenant_models.py
- a5b6c7d8e9f0_add_kol_region_columns.py
- c4665648bc07_add_metadata_fields_to_kol_groups_and_.py
- b5c6d7e8f9a0_kol_hcp_link.py
- d8e9f0a1b2c3_kol_hcp_unify.py

HCP Intel: all migrations touching hcps, hcp_signals, and related tables (grep hcp_intel / table names in backend/migrations/versions/).

5. Background jobs (Lambda catalog)

From sync/README.md and backend/src/jobs/scheduler.py:

Job	Schedule	Purpose
hcp_intel_poll	30m	Feed orchestrator - PubMed, ClinicalTrials, Google News, etc.
hcp_intel_openalex_backfill	Sun 03:30 UTC	Publication signals for linked HCPs
kol_hcp_matcher	On-demand or daily	Resolve kols.hcp_npi
post_tagging	12h	Keyword + KOL name tags on clips
platform_sync	12h	YouTube/LinkedIn/X ingest
ai_summaries	12h	Clip/shoot AI descriptions

Drop: kol_cache_warm - CHT owns Redis (cht:kol-network:{hash(params)}).

Additional on-demand admin ops (triggered from studio API, not scheduled): signal purge, ensure-plumbing, resolve-unmatched, re-ingest-webinars, CMS Part D load, Open Payments sync, NIH sync.

6. Frontend integration (new build)

6.1 Public API contract - consumer /kol-network

Document these shapes for the new frontend team (extend cht-public-api-contract.md as needed):

GET /api/public/kols - list + facets

Field	Type	Notes
items[].id, slug, name, title, specialty, institution, bio, photo_url	string	Core profile
items[].region, region_label	string	Filter chips
items[].shoot_count, first_appeared_at, is_new	int / datetime / bool	Engagement signals
regions[], institutions[]	facets	Directory filters
Query: region, institution, q, new_only, limit, offset		

GET /api/public/kols/{slug}/publications - publication list (empty array OK).

Enrichment fields to add before new UI ships (currently mocked in legacy dol-network.ts):

Field	Source table / join
npi	kols.hcp_npi -> hcps.npi
education	hcps.enrichment_data or new column

linkedinUrl, twitterUrl, webUrl	hcps.enrichment_data
aiBrief	hcp_ai_briefs
openPayments summary	open_payments_records aggregate
publicationsApprox	count from hcp_signals

Either extend PublicKOL on the detail endpoint or add GET /api/public/kols/{slug}/enrichment - decide before frontend build starts.

6.2 Admin studio API - new admin UI

All routes under /api/admin/studio/hcp-intel/* and /api/admin/studio/kols/* with Cognito session JWT + chm-admin / chm-editor / chm-viewer group checks. Export OpenAPI from FastAPI for frontend code generation.

6.3 CHT registration path (server-to-server - still required)

Independent of consumer UI redo. CHT backend must call Content Hub on signup/profile update:

Caller	Target
Registration / profile sync	POST {CONTENTHUB}/api/public/hcp/upsert

Update env: MEDIAHUB_BASE_URL -> CONTENTHUB_BASE_URL in cht-platform-tool (or successor backend).

Optional retirement: cht-platform-tool/backend/src/modules/kol-network/ proxy module - drop if new frontend or BFF calls Content Hub public API directly with appropriate auth. Registration upsert stays on CHT backend regardless.

6.4 Consumer screens - reference checklist (new frontend)

Screen	Public API
KOL directory (region/state filters)	GET /kols
KOL profile	GET /kols/{slug} + publications + enrichment
Region detail	GET /kols?region=...

Legacy reference: cht-platform-tool/frontend/src/pages/public/DolNetwork.tsx, KolProfilePage.tsx, DolRegionDetail.tsx.

7. Cutover phases

Phase 1 - Data + producer API (backend only)

- pg_dump all KOL + HCP Intel tables (see Section 4) into Content Hub Aurora
- Upload headshots to S3; rewrite kols.photo_url
- Port backend/src/ producer code (public KOL + HCP upsert)
- Port full hcp_intel/ package
- Deploy contenthub-api to ECS; smoke test public endpoints
- Wire Lambda jobs (hcp_intel_poll, openalex_backfill, post_tagging)
- Publish API contract doc / OpenAPI for frontend team (Section 6)

Phase 2 - Server integration

- Point CHT registration sync to Content Hub (POST /hcp/upsert)
- Verify HCP upsert round-trip from CHT signup flow
- Extend PublicKOL (or enrichment endpoint) with fields in Section 6.1
- Smoke test: `curl -H "X-API-Key: $KEY" "$BASE/kols?limit=1"`

No consumer UI verification in this phase - new frontend not built yet.

Phase 3 - Admin studio API (backend only)

- Port /api/admin/studio/hcp-intel/* routes with Cognito JWT middleware
- Port /api/admin/studio/kols/* routes
- Export OpenAPI for admin studio surface
- Retire MediaHub /api/hcp-intel/* routes once studio API is live

New admin UI is a separate frontend workstream - consumes studio API when ready.

Phase 4 - MediaHub decommission (KOL + HCP Intel)

- Remove KOL/HCP Intel routes from MediaHub legacy/main.py and src/main.py
- Drop KOL + HCP Intel tables from MediaHub Postgres (after validation window)
- Delete static asset directories from MediaHub repo
- Retire legacy frontend code in MediaHub + cht-platform-tool (when new UI ships)
- Update docs: no MediaHub references for KOL or HCP Intel

8. MediaHub exit criteria (KOL + HCP Intel)

MediaHub backend/data is clear when all of the following are true:

- No kols, kol_groups, kol_group_members, or HCP Intel tables in MediaHub DB
- No /api/public/kols* or /api/public/hcp/upsert on MediaHub hostnames
- No /api/hcp-intel/* routes on MediaHub
- No /static/kol-headshots/ mount or files in MediaHub deploy artifact
- CHT registration sync points at Content Hub (not MediaHub)
- Content Hub public + studio APIs pass smoke tests

Legacy frontend retirement (separate from backend exit criteria - complete when new UI ships):

- MediaHub /dashboard/hcps/* and KOL group pages removed or unreachable
- cht-platform-tool KOL pages and dol-network.ts retired
- New consumer + admin UIs live against Content Hub APIs

9. Risk notes

Risk	Mitigation
Large HCP Intel surface (~60 routes, 22 tables, 140+ source files)	Port src/hcp_intel/ as a unit; keep legacy router as thin re-export during transition
Publications empty until OpenAlex backfill runs	Run backfill immediately after data port; matcher job links KOLs to NPIs first
Report generator still references local kol_photos/	Point report automation at Content Hub CDN URLs before Phase 4
Client dashboard KOL group counts	Add read-only Content Hub API for analytics embed, or drop counts from MediaHub UI
New frontend blocked on enrichment fields	Ship Section 6.1 enrichment on API before frontend build; do not recreate static mocks
Frontend team starts before OpenAPI exists	Block frontend kickoff until Phase 1 publishes contract doc

10. File manifest (backend port targets)

Port to cht-content-hub:

```

chm-mediashub/backend/src/
|---- models/kol.py
|---- schemas/public.py          (KOL + HCP upsert schemas)
|---- utils/public.py            (slug helpers)
|---- services/kol_regions.py
|---- services/hcp_upsert.py
|---- public/router.py           (KOL + HCP upsert handlers)
`---- hcp_intel/                 (entire package - 30+ modules, tests)

chm-mediashub/backend/legacy/
|---- routers/hcp_intel.py       (admin API -> /api/admin/studio/hcp-intel/*)
|---- routers/clients.py         (KOL group routes -> /api/admin/studio/kols/*)
|---- static/kol-headshots/      (29 PNG -> S3)
`---- report_automation/assets/kol_photos/ (~15 JPG -> S3)

```

Reference only - do not port (frontend redo):

```

chm-mediashub/frontend/src/app/dashboard/hcps/      (12 pages - screen checklist Section 3.3)
chm-mediashub/frontend/.../KOLGroupCard.tsx
chm-mediashub/frontend/.../clients/.../groups/

cht-platform-tool/frontend/src/pages/public/Dol*.tsx, KolProfilePage.tsx
cht-platform-tool/frontend/src/data/dol-network.ts
cht-platform-tool/frontend/src/hooks/useKolDirectory.ts
cht-platform-tool/backend/src/modules/kol-network/  (optional retire)

```